Constituency Tests for Concerned Representation in Minimal Agents

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Abstract

Arc 1 of the maintained-concern program showed that a minimal homeostatic agent can detect boundary staleness, allocate costly null probes, saturate after identification, and re-engage after regime shifts. It also found a precise ceiling: under a shared-head, null-only intervention regime, better probe policy no longer closes role-specific mediated identifiability. This paper opens Arc 2A by asking whether the next intervention problem is syntactic: can a concerned agent discover which parts of a world belong together causally?

We introduce the **Concerned Shape Grammar**, a symbolic benchmark inspired by constituency tests from cognitive work on geometric shape syntax. A visible six-part shape can have the same surface but different hidden parse trees. Causal roles such as shield, poison, repair, core, food, and trap interact only when they are bound inside the same constituent. The agent must decide whether to pay for an intervention that reveals the parse, and it must avoid probing low-concern ambiguity that does not affect viability.

In a 200-trial deterministic design pilot and a 5,000-trial Modal multi-seed sweep, the `concerned_syntax` selector is the only selector that passes the full gate. In the Modal sweep, it reaches high-concern parse accuracy 1.000, action accuracy 1.000, high-concern probe rate 1.000, low-concern probe rate 0.000, and gate pass rate 1.000. Flat valence, compression-only parsing, and null policies fail because they do not make constituency knowable. An uncertainty-only selector recovers the parse but fails the no-restless-inquiry gate by probing every low-concern ambiguity. The result is not a claim about neural agents yet. It is an accepted Phase 2A benchmark surface: **reward is not syntax, compression is not syntax, and uncertainty reduction is not concerned inquiry.**

1. Why Arc 2A Exists

The Metric Stack of Concern ended with a positive mechanism and a ceiling. The positive mechanism was a detect-allocate-saturate-re-engage cycle for self/world attribution in minimal homeostatic agents. The ceiling was equally important: under null-only intervention and shared mediated heads, the agent could predict total world response while failing to identify role-specific mediated components.

That ceiling has two sides. Arc 2B studies the body side: what architectures can express the required distinctions? Arc 2A studies the intervention side: what actions make the world's hidden grammar visible?

The motivation comes from a nearby but distinct empirical result. Revencu, Pajot, and Dehaene (2026) argue that adult geometric-shape representations show syntactic structure. Their key methodological move is not merely to show that shape complexity predicts behavior. They replace compression proxies with constituency tests: structural ambiguity, subtree facilitation, and syntactic movement. They also report that current neural networks can partially solve match/deviant tasks without showing the same syntactic effects.

This paper imports that methodological lesson into the concern program. The claim is not that minimal agents have human visual syntax. The claim is that the Metric Stack needs an analogous gate:

> Performance is not constituency. Compression is not constituency. Uncertainty reduction is not concerned inquiry.

2. Benchmark

Each trial has a six-part visible shape. The visible role sequence is fixed, but two hidden parse trees can organize the same surface differently. A parse is a pair of high-level constituents, each containing three leaves.

Examples of parse candidates:

```
repeat_concat     = (0,1,2) | (3,4,5)
hooked_repeat     = (0,1,3) | (2,4,5)
alternating_bind = (0,2,4) | (1,3,5)
edge_core         = (0,4,5) | (1,2,3)
```

Causal roles interact only when they belong to the same constituent:

Role pair	Causal rule
shield + poison	shield reduces poison damage only within a subtree
repair + core	repair reduces core damage only within a subtree
food + trap	trap contaminates food only within a subtree
signal + ornament	structural ambiguity with no viability consequence

The low-concern case is essential. Without it, a generic uncertainty reducer could look successful simply by probing every ambiguity.

3. Interventions

The intervention language contains five operations:

Intervention	Purpose
`null`	no information, no cost
`pair_probe`	test whether the causal role pair is in the same subtree
`distractor_pair_probe`	matched non-informative pair probe
`high_constituent_move`	test which high-level group moves as a unit
`role_ablation`	observe the viability signature of the causal constituent

The central selector, `concerned_syntax`, chooses an intervention by expected parse information weighted by the viability gap, minus intervention cost. It has a hard no-restless-inquiry threshold: if the parse gap is below 0.10, it chooses `null`.

4. Selectors

Selector	What it tests
`null_policy`	behavior without inquiry
`flat_valence`	roles without constituency
`compression_proxy`	shortest parse without intervention
`uncertainty_only`	information gain without concern
`concerned_syntax`	information gain gated by viability relevance

This is the anti-cheat surface. A selector can be good at action while failing parse. It can recover parse while probing too much. It can choose a compressed parse while missing the causally relevant one.

5. Pilot Result

Local command:

```
python3 -m experiments.concerned_syntax.benchmark \
  --trials 200 --seed 20260616 \
  --out artifacts/concerned_syntax/pilot.json \
  --report experiments/concerned_syntax/results/pilot_2026_06_16.md
```

Summary:

Selector	Parse high	Action	Subtree	High probe	Low probe	Mean regret	Gate
compression_proxy	0.523	0.890	0.550	0.000	0.000	0.055	fail
concerned_syntax	1.000	1.000	0.805	1.000	0.000	0.001	PASS
flat_valence	0.000	0.920	0.540	0.000	0.000	0.074	fail
null_policy	0.523	0.890	0.550	0.000	0.000	0.055	fail
uncertainty_only	1.000	1.000	1.000	1.000	1.000	0.000	fail

The result has the desired diagnostic pattern:

1. `flat_valence` has high action accuracy but fails parse and subtree gates.
2. `compression_proxy` sometimes guesses the right parse, but does not intervene and fails the mechanistic gate.
3. `uncertainty_only` recovers parse perfectly but probes all low-concern ambiguity, failing no-restless-inquiry.
4. `concerned_syntax` probes every high-concern ambiguity, avoids low-concern probing, and preserves action.

6. Discovery-Regime Status

This is a regime transition relative to Arc 1. Arc 1 represented concern as viability prediction, valence, self/

world attribution, probe value, and component identifiability. Arc 2A adds a new artifact type: **causal constituency under concern**.

That makes the result discovery-leaning, but still benchmark-level. The pilot does not show that a neural agent has learned syntax. It shows that the repo now has a task surface on which syntax, compression, uncertainty, and concern can dissociate.

7. Modal Multi-Seed Sweep

The multi-seed sweep was run remotely through Modal:

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/concerned_syntax/modal_concerned_syntax_sweep.py \
  --trials 1000
```

The sweep used five seeds and 1,000 shape trials per seed. Raw JSON remains local under `artifacts/concerned_syntax/`; the public report is `experiments/concerned_syntax/results/modal_sweep_2026_06_16.md`.

Summary:

Selector	Parse high	Action	Subtree	High probe	Low probe	Mean regret	Gate pass rate
compression_proxy	0.560	0.891	0.583	0.000	0.000	0.048	0.000
concerned_syntax	1.000	1.000	0.808	1.000	0.000	0.003	1.000
flat_valence	0.000	0.876	0.503	0.000	0.000	0.066	0.000
null_policy	0.560	0.891	0.583	0.000	0.000	0.048	0.000
uncertainty_only	1.000	1.000	1.000	1.000	1.000	0.000	0.000

This replicates the design-pilot pattern across seeds. The key result is not that `concerned_syntax` has the best reward; `uncertainty_only` has zero regret too. The key result is that only concern-weighted syntax passes both the positive inquiry gate and the no-restless-inquiry gate.

The next version should add learned agents:

- tree-structured model
- flat MLP baseline
- object-slot baseline
- learned intervention policy
- role/parse held-out generalization
- neural anti-cheat probes for parse, subtree, and intervention usefulness

8. Limitations

The current benchmark is symbolic. It does not test pixels, learned perception, continuous control, or human subjects. The intervention language is provided, not invented from raw motor primitives. The parser candidates are small and known to the evaluator. The point of this first paper is to define the acceptance surface before larger compute.

The most important limitation is also the next step: the agent should learn the intervention language and parse representation, not merely select among hand-defined probes.

9. Conclusion

Arc 2A inserts a new layer into the maintained-concern ladder:

```
difference -> geometry -> syntax -> salience -> valence
          -> action -> attribution -> maintenance
```

The pilot supports a narrow methodological claim: concerned syntax needs its own tests. Reward, compression, uncertainty, and action accuracy can all dissociate from causal constituency. The next experiments can now ask whether learned agents and evolved bodies pass those tests without cheating.

References

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